

Bound the integrality gap of time-indexed LP for bipartite AND/OR scheduling

ChatGPT (gpt-5.6-sol)*

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Abstract

We determine the exact supremal integrality gap of the time-indexed linear program introduced by Happach and Schulz for single-machine scheduling with AND/OR precedence constraints and processing times in $\{0, 1\}$. If

$$\Delta = \max \left\{ 1, \max_{b \in B} |P(b)| \right\}$$

denotes the maximum number of OR-predecessors of a job, then the gap is exactly 2Δ .

The upper bound follows from the random α -point argument of Happach and Schulz; we give an endpoint-safe proof that does not invoke a precedence constraint at the unavailable index zero. For the lower bound, we construct a finite family based on all Δ -subsets of a ground set: its integer optimum is a hitting-set number, while its fractional optimum is determined exactly by a cumulative-capacity argument. We also prove that a separate grouped-star instance admits an optimal fractional solution and a legal tie resolution under which the α -point algorithm has an approximation ratio tending to 2Δ . This last statement concerns the existence of a bad legal execution, not every optimal LP solution or every tie-breaking rule.

1 The open problem

This paper addresses an open problem posed by Felix Happach, Andreas S. Schulz in *Approximation Algorithms and Linear Programming Relaxations for Scheduling Problems Related to Min-Sum Set Cover* [1] (Mathematics of Operations Research, 2023), Section 7 (Conclusion), page 22 (arXiv v1 pagination). The website catalogue entry is problem 78 (w4377821221_p0) of the [Open Problems in Operations Research](#) collection.

*Generated by the `base_solver_non_interactive` (MOR sweep) pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.10 repair write-up; run `run_20260821_195826_583ab9fd`, `sweep-audit-approved`). Independent verifier assessment: REJECTED with progress score 3/3 (full solution); score 3/3 but not approved. Maintained by the AI in Mathematics & Operations Research project.

Determine the best (smallest) constant $g \geq 1$ such that, for every instance of $1 \mid \text{ao-prec} = A \uplus B \mid \sum_{j \in N} w_j C_j$ and for the corresponding time-indexed linear programming relaxation described above, the ratio

$$\frac{\text{optimal objective value of an optimal integer schedule}}{\text{optimal objective value of the time-indexed LP relaxation}}$$

is at most g . Equivalently, bound the integrality gap of the time-indexed LP relaxation for this problem class (as a function of parameters such as Δ , or by a universal constant if possible), and in particular decide whether the approximation-factor analyses obtained via random α -point rounding are tight in the worst case.

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

2 Introduction

Happach and Schulz study single-machine scheduling with a job set $N = A \dot{\cup} B$. AND-precedence arcs lie within A or within B , while OR-precedence arcs go from A to B . For processing times $p_j \in \{0, 1\}$, they introduce a time-indexed relaxation and prove a 2Δ LP-relative approximation bound using randomized α -point scheduling; see [2]. Their concluding discussion asks for stronger information about the integrality gap of this relaxation.

This note answers that question for the specific 0/1 time-indexed relaxation. The result is a supremal statement: for every positive integer d , all finite instances with maximum OR-in-degree at most d have gap at most $2d$, and a sequence of finite instances with maximum OR-in-degree exactly d has ratios converging to $2d$.

Two scope restrictions are important. First, the construction uses zero-processing-time jobs and does not prove the same lower bound when every job has unit processing time. Second, the result concerns the time-indexed $p_j \in \{0, 1\}$ relaxation, not the geometric interval relaxation for arbitrary processing times.

3 Model and time-indexed relaxation

Let $N = A \dot{\cup} B$ be a finite set of jobs. Each job j has processing time $p_j \in \{0, 1\}$ and a nonnegative weight w_j . The precedence arcs are partitioned as

$$E_{\wedge} \subseteq (A \times A) \cup (B \times B), \quad E_{\vee} \subseteq A \times B.$$

For $b \in B$, let

$$P(b) := \{a \in A : (a, b) \in E_{\vee}\}.$$

A feasible schedule must place every AND-predecessor of a job before that job, and it must place at least one member of $P(b)$ before every job b with $P(b) \neq \emptyset$. Zero-time jobs may be ordered sequentially at the same clock time.

We assume that the AND-precedence relation is acyclic and that

$$T := \sum_{j \in N} p_j \geq 1.$$

Write $[k] = \{1, \dots, k\}$ for $k \geq 1$ and $[0] = \emptyset$. Following the normalization in [2], we assume initially that $w_a = 0$ for every $a \in A$.

The variable x_{jt} denotes the fraction of job j that completes at time $t \in [T]$. Define

$$F_j(t) \doteq \sum_{s=1}^t x_{js}, \quad F_j(0) \doteq 0.$$

The relaxation is

$$\min \quad \sum_{b \in B} w_b \sum_{t=1}^T t x_{bt} \tag{1a}$$

$$\text{s.t.} \quad \sum_{t=1}^T x_{jt} = 1 \quad j \in N, \tag{1b}$$

$$\sum_{j \in N} \sum_{s=t-p_j+1}^t x_{js} \leq 1 \quad t \in [T], \tag{1c}$$

$$F_b(t + p_b) \leq \sum_{a \in P(b)} F_a(t) \quad b \in B, P(b) \neq \emptyset, t \in [T - p_b], \tag{1d}$$

$$F_j(t + p_j) \leq F_i(t) \quad (i, j) \in E_\wedge, t \in [T - p_j], \tag{1e}$$

$$x_{jt} \geq 0 \quad j \in N, t \in [T]. \tag{1f}$$

The inner sum in (1c) is empty when $p_j = 0$; consequently, only unit-time jobs consume capacity.

For a feasible fractional solution, define

$$\overline{C}_j \doteq \sum_{t=1}^T t x_{jt}, \quad Q_j(\gamma) \doteq \min\{t \in [T] : F_j(t) \geq \gamma\}, \quad 0 < \gamma \leq 1.$$

Lemma 1 (Tail and quantile identities). *For every job j ,*

$$\overline{C}_j = \sum_{t=0}^{T-1} (1 - F_j(t)) = \int_0^1 Q_j(\gamma) d\gamma.$$

Proof. The first equality follows by interchanging the finite sums:

$$\sum_{t=0}^{T-1} (1 - F_j(t)) = \sum_{s=1}^T s x_{js}.$$

Except at finitely many endpoints, $Q_j(\gamma) = t$ for $\gamma \in (F_j(t-1), F_j(t)]$. Therefore

$$\int_0^1 Q_j(\gamma) d\gamma = \sum_{t=1}^T t (F_j(t) - F_j(t-1)) = \sum_{t=1}^T t x_{jt}. \quad \square$$

Define

$$\Delta(I) \doteq \max \left\{ 1, \max_{b \in B} |P(b)| \right\},$$

where the maximum over an empty set is zero. For an integer $d \geq 1$, let

$$g(d) = \sup \left\{ \frac{\text{OPT}(I)}{\text{LP}(I)} : \begin{array}{l} I \text{ is a finite instance of (1a)–(1f),} \\ \Delta(I) \leq d, \quad \text{LP}(I) > 0 \end{array} \right\}.$$

Our main result is the following.

Theorem 1. *For every positive integer d ,*

$$\boxed{g(d) = 2d}.$$

Thus the supremal integrality gap of LP (1a)–(1f) is 2Δ . In particular, no constant independent of Δ can bound the gap.

3.1 Weights on jobs in A

The normalization $w_a = 0$ is harmless, but preservation of the fractional optimum requires an argument.

Lemma 2 (Weight-copy reduction). *Suppose the natural unreduced objective assigns a positive weight w_a to a job $a \in A$. Replace that weight by a zero-time job $c_a \in B$ with*

$$p_{c_a} = 0, \quad w_{c_a} = w_a, \quad P(c_a) = \{a\},$$

and set the weight of a to zero. This transformation preserves both the integer optimum and the fractional optimum. It does not increase $\Delta(I)$ under the convention $\Delta(I) \geq 1$.

Proof. In an integer schedule, c_a cannot occur before a ; hence $C_{c_a} \geq C_a$. Conversely, c_a can be placed immediately after a without consuming processing time, giving $C_{c_a} = C_a$. Restricting and extending schedules therefore proves that the two integer optima are equal.

In the transformed LP, constraint (1d) gives

$$F_{c_a}(t) \leq F_a(t) \quad (t \in [T]).$$

By Lemma 1,

$$\overline{C}_{c_a} = \sum_{t=0}^{T-1} (1 - F_{c_a}(t)) \geq \sum_{t=0}^{T-1} (1 - F_a(t)) = \overline{C}_a.$$

Thus restricting any solution of the transformed LP to the original jobs cannot increase the corresponding original objective. In the other direction, any fractional solution of the original LP can be extended by setting $x_{c_a t} = x_{at}$ for every t . This consumes no capacity and satisfies (1d) with equality. Hence the two fractional optima are equal. The new OR-predecessor set has cardinality one, so $\Delta(I)$ does not increase. $\square$

4 An endpoint-safe 2Δ upper bound

Summing (1c) over the slots $1, \dots, t$ yields the cumulative capacity inequality

$$\sum_{j \in N} p_j F_j(t) \leq t, \quad t = 0, \dots, T. \tag{1}$$

Indeed, a job with $p_j = 1$ contributes $F_j(t)$, while jobs with $p_j = 0$ contribute nothing.

Fix $\beta \in (0, 1]$ and put

$$\alpha = \frac{\beta}{\Delta}.$$

Assign to each job the key

$$K_j = \begin{cases} Q_j(\alpha), & j \in A, \\ Q_j(\beta), & j \in B. \end{cases}$$

Order the jobs by nondecreasing key. Among jobs with the same key, place jobs in A before jobs in B , and within each class use a topological order of the AND-arcs whose endpoints share that key.

Lemma 3 (Feasibility of the quantile order). *The resulting order is a feasible schedule for every $\beta \in (0, 1]$.*

Proof. We first consider an AND-arc $(i, j) \in E_\wedge$. Its endpoints belong to the same class and therefore use the same threshold, say γ . Let

$$q = Q_j(\gamma).$$

If $q \leq T - p_j$, apply (1e) at the available index $t = q$:

$$F_i(q) \geq F_j(q + p_j) \geq F_j(q) \geq \gamma.$$

Thus $Q_i(\gamma) \leq q$.

Suppose next that $q > T - p_j$ and $T - p_j \geq 1$. Applying (1e) at its last available index gives

$$F_i(T - p_j) \geq F_j(T) = 1,$$

and hence

$$Q_i(\gamma) \leq T - p_j < q.$$

The remaining case $T - p_j = 0$ forces $T = p_j = q = 1$. By (1b), every job's distribution is supported at the single time index 1, so $Q_i(\gamma) = Q_j(\gamma) = 1$. The prescribed topological tie-breaking then places i before j . Thus every AND-arc is respected.

Now fix $b \in B$ with $P(b) \neq \emptyset$ and set

$$q = Q_b(\beta).$$

If $q \leq T - p_b$, apply (1d) at $t = q$:

$$\beta \leq F_b(q) \leq F_b(q + p_b) \leq \sum_{a \in P(b)} F_a(q).$$

Therefore some $a \in P(b)$ satisfies

$$F_a(q) \geq \frac{\beta}{|P(b)|} \geq \frac{\beta}{\Delta} = \alpha,$$

and consequently $Q_a(\alpha) \leq q$.

If $q > T - p_b$ and $T - p_b \geq 1$, apply (1d) at its final available index:

$$1 = F_b(T) \leq \sum_{a \in P(b)} F_a(T - p_b).$$

Some $a \in P(b)$ then satisfies

$$F_a(T - p_b) \geq \frac{1}{|P(b)|} \geq \frac{1}{\Delta} \geq \frac{\beta}{\Delta} = \alpha,$$

and hence

$$Q_a(\alpha) \leq T - p_b < q.$$

Finally, if $T - p_b = 0$, then $T = p_b = q = 1$, and every positive quantile of every job equals 1. The rule placing A before B at equal keys puts each $a \in P(b)$ before b .

Thus every job b with an OR-requirement has at least one OR-predecessor before it, and every AND-arc is respected. $\square$

Remark. The proof deliberately applies the precedence constraints at the index q or at their last available index $T - p_j$. Substituting $q - p_j$ would be invalid when $p_j = q = 1$, because the LP as stated has no constraint indexed by zero.

Proposition 1 (Upper bound). *For every finite instance I ,*

$$\text{OPT}(I) \leq 2\Delta(I) \text{ LP}(I).$$

Proof. Fix $b \in B$ and let

$$q = Q_b(\beta).$$

Let $R_b(\beta)$ denote the prefix of the schedule ending at b . Every $i \in R_b(\beta)$ has $K_i \leq q$, and hence

$$F_i(q) \geq \begin{cases} \alpha, & i \in A, \\ \beta \geq \alpha, & i \in B. \end{cases}$$

The list schedule contains no idle time, so

$$C_b(\beta) = \sum_{i \in R_b(\beta)} p_i.$$

Using (1),

$$\begin{aligned} \alpha C_b(\beta) &= \sum_{i \in R_b(\beta)} \alpha p_i \\ &\leq \sum_{i \in R_b(\beta)} p_i F_i(q) \\ &\leq \sum_{i \in N} p_i F_i(q) \\ &\leq q. \end{aligned}$$

Since $\alpha = \beta/\Delta$,

$$C_b(\beta) \leq \frac{\Delta}{\beta} Q_b(\beta). \tag{2}$$

Draw β from $(0, 1]$ with density $f(\beta) = 2\beta$. By Lemma 1,

$$\begin{aligned}\mathbb{E}[C_b(\beta)] &\leq \int_0^1 2\beta \frac{\Delta}{\beta} Q_b(\beta) d\beta \\ &= 2\Delta \int_0^1 Q_b(\beta) d\beta \\ &= 2\Delta \bar{C}_b.\end{aligned}$$

Only jobs in B carry positive weight, and hence

$$\mathbb{E} \left[\sum_{b \in B} w_b C_b(\beta) \right] \leq 2\Delta \sum_{b \in B} w_b \bar{C}_b.$$

By Lemma 3, every realization yields a feasible schedule; therefore some realization costs no more than this expectation, which proves the proposition. The case of general weights on A follows from Lemma 2. $\square$

5 A matching lower-bound family

Fix integers $d, q \geq 1$ and set

$$m := dq.$$

Construct the following instance.

- Create jobs $a_1, \dots, a_m \in A$ with

$$p_{a_i} = 1, \quad w_{a_i} = 0.$$

- For every d -subset $S \subseteq [m]$, create a job $b_S \in B$ with

$$p_{b_S} = w_{b_S} = 0, \quad P(b_S) = \{a_i : i \in S\}.$$

- Create a job $z \in B$ with

$$p_z = 0, \quad w_z = 1,$$

and add the AND-arcs

$$(b_S, z) \in E_\wedge \quad \text{for every } S \subseteq [m], |S| = d.$$

This yields a finite instance with

$$\Delta = d, \quad T = m.$$

Proposition 2 (Integer optimum). *For this instance,*

$$\text{OPT}(I_{d,q}) = m - d + 1 = dq - d + 1.$$

Proof. Consider any feasible schedule and let $H \subseteq [m]$ be the set of indices of the A -jobs placed before z in the schedule order. Every b_S must occur before z because $(b_S, z) \in E_\wedge$. For b_S to occur, at least one a_i with $i \in S$ must already have occurred. Therefore

$$H \cap S \neq \emptyset \quad \text{for every } S \subseteq [m] \text{ with } |S| = d.$$

The complement $[m] \setminus H$ thus contains no d -subset, and hence

$$|H| \geq m - d + 1.$$

Since the jobs indexed by H have unit processing time,

$$C_z \geq |H| \geq m - d + 1.$$

Conversely, any subset $H \subseteq [m]$ of size $m - d + 1$ intersects every d -subset of $[m]$. Schedule the jobs indexed by H first, then all zero-time jobs b_S , then z , and finally the remaining A -jobs. This schedule is feasible and gives

$$C_z = m - d + 1. \quad \square$$

We next exhibit a feasible fractional solution. For every $i \in [m]$, set

$$x_{a_i, t} = \frac{1}{m} \quad (t = 1, \dots, m).$$

For every d -subset S and for z , set

$$x_{b_S, t} = x_{z, t} = \begin{cases} 1/q, & 1 \leq t \leq q, \\ 0, & q < t \leq m. \end{cases}$$

The m unit-time jobs contribute exactly one unit of capacity in every slot, and all other jobs have processing time zero. Moreover,

$$F_{a_i}(t) = \frac{t}{m} \quad (0 \leq t \leq m),$$

while

$$F_{b_S}(t) = F_z(t) = \min \left\{ \frac{t}{q}, 1 \right\}.$$

For every S with $|S| = d$,

$$\sum_{i \in S} F_{a_i}(t) = \frac{dt}{m} = \frac{t}{q} \geq F_{b_S}(t),$$

and $F_z(t) = F_{b_S}(t)$. Thus all OR- and AND-constraints hold.

The value of this solution is

$$\overline{C}_z = \frac{1}{q} \sum_{t=1}^q t = \frac{q+1}{2}. \quad (3)$$

Proposition 3 (Fractional optimum). *For the lower-bound instance,*

$$\text{LP}(I_{d,q}) = \frac{q+1}{2}.$$

Proof. Consider any feasible fractional solution and fix $t \in \{0, \dots, m\}$. Put

$$y_i \doteq F_{a_i}(t).$$

For every d -subset S , the AND- and OR-constraints imply

$$F_z(t) \leq F_{b_S}(t) \leq \sum_{i \in S} y_i.$$

Consequently

$$F_z(t) \leq \min_{\substack{S \subseteq [m] \\ |S|=d}} \sum_{i \in S} y_i.$$

The minimum equals the sum of the d smallest coordinates of y ; their average is at most the average of all m coordinates, so

$$\min_{\substack{S \subseteq [m] \\ |S|=d}} \sum_{i \in S} y_i \leq \frac{d}{m} \sum_{i=1}^m y_i.$$

The cumulative capacity inequality gives

$$\sum_{i=1}^m y_i = \sum_{i=1}^m F_{a_i}(t) \leq t.$$

Since $m = dq$,

$$F_z(t) \leq \frac{dt}{m} = \frac{t}{q}. \tag{4}$$

By the tail identity of Lemma 1,

$$\begin{aligned} \overline{C}_z &= \sum_{t=0}^{m-1} (1 - F_z(t)) \\ &\geq \sum_{t=0}^{q-1} \left(1 - \frac{t}{q}\right) \\ &= \frac{q+1}{2}. \end{aligned}$$

Together with (3), this proves optimality. $\square$

Combining Propositions 2 and 3 yields

$$\frac{\text{OPT}(I_{d,q})}{\text{LP}(I_{d,q})} = \frac{2(dq - d + 1)}{q + 1} = 2d - \frac{4d - 2}{q + 1}, \tag{5}$$

and thus

$$\lim_{q \rightarrow \infty} \frac{\text{OPT}(I_{d,q})}{\text{LP}(I_{d,q})} = 2d.$$

Proof of Theorem 1. For every instance with $\Delta(I) \leq d$, Proposition 1 gives

$$\frac{\text{OPT}(I)}{\text{LP}(I)} \leq 2\Delta(I) \leq 2d.$$

Hence $g(d) \leq 2d$.

For every $q \geq 1$, the instance $I_{d,q}$ is finite and has $\Delta(I_{d,q}) = d$. Equation (5) shows that its ratio converges to $2d$. Therefore $g(d) \geq 2d$, proving $g(d) = 2d$. $\square$

Remark. The construction contains

$$m + \binom{m}{d} + 1$$

jobs. This causes no difficulty for an integrality-gap lower bound, since each member of the sequence is finite. For fixed d , its size is moreover polynomial in q .

6 A bad legal execution of the α -point algorithm

The lower-bound family shows that the LP-relative factor $2d$ cannot be improved: every feasible schedule on that family costs at least $\text{OPT}(I_{d,q})$. It does not by itself imply that the particular α -point algorithm has approximation ratio $2d$ relative to the integer optimum.

The following separate construction shows that the algorithm as stated admits legal executions whose actual approximation ratios tend to $2d$. The claim is intentionally existential with respect to both the chosen optimal LP solution and the resolution of permitted ties.

Proposition 4. *For every $d \geq 1$ and $n \geq 2$, there is an instance with $\Delta = d$, an optimal LP solution, and a legal tie-breaking of the α -point order for which*

$$\frac{\text{ALG}}{\text{OPT}} = \frac{2dn}{n+2}.$$

In particular, these ratios tend to $2d$ as $n \rightarrow \infty$.

Proof. Create n groups

$$G_0, G_1, \dots, G_{n-1},$$

where G_0 is the center group and the remaining groups are leaves. Each G_v consists of d unit-time, zero-weight jobs in A .

For every $v \in \{0, \dots, n-1\}$, create a zero-time, zero-weight helper $u_v \in B$ with

$$P(u_v) = G_v.$$

For each leaf index $i \in \{1, \dots, n-1\}$, create a zero-time job $e_i \in B$ of weight one and add the two AND-arcs

$$(u_0, e_i), \quad (u_i, e_i).$$

The time horizon is $T = nd$, and $\Delta = d$.

Integer optimum. Let τ_v denote the completion time of the first unit-time job selected from G_v . Once the necessary zero-time helpers are inserted, the earliest possible completion time of e_i is

$$C_{e_i} = \max\{\tau_0, \tau_i\}.$$

The n values $\tau_0, \dots, \tau_{n-1}$ are distinct completion times of unit-time jobs. Write their increasing order as

$$\sigma_1 < \sigma_2 < \dots < \sigma_n.$$

Then $\sigma_k \geq k$. Suppose the center activation τ_0 has rank r . Each of the $r - 1$ leaves activated before the center contributes at least r , while the leaves of ranks $r + 1, \dots, n$ contribute at least their ranks. Therefore

$$\sum_{i=1}^{n-1} C_{e_i} \geq \phi(r) = r(r-1) + \sum_{k=r+1}^n k.$$

Since

$$\phi(r+1) - \phi(r) = r - 1,$$

the minimum is attained at $r = 1$ or $r = 2$, and its value is

$$\sum_{k=2}^n k = \frac{(n-1)(n+2)}{2}.$$

This value is attainable by processing one center job first and then one job from each leaf group, inserting each available helper and weighted job immediately. Hence

$$\text{OPT} = \frac{(n-1)(n+2)}{2}. \quad (6)$$

A lower bound on every fractional solution. For $t = 0, \dots, T$, define

$$Y_v(t) = \sum_{a \in G_v} F_a(t), \quad f_i(t) = F_{e_i}(t), \quad S(t) = \sum_{i=1}^{n-1} f_i(t).$$

The OR-constraints for the helpers and the AND-constraints for the weighted jobs give

$$Y_0(t) \geq F_{u_0}(t) \geq \max_i f_i(t)$$

and

$$Y_i(t) \geq F_{u_i}(t) \geq f_i(t) \quad (i = 1, \dots, n-1).$$

The cumulative unit-job capacity implies

$$Y_0(t) + \sum_{i=1}^{n-1} Y_i(t) \leq t.$$

Consequently

$$t \geq \max_i f_i(t) + S(t) \geq \frac{S(t)}{n-1} + S(t) = \frac{n}{n-1} S(t),$$

and thus

$$S(t) \leq \frac{n-1}{n} t. \quad (7)$$

Using the tail identity and retaining only the terms $t = 0, \dots, n-1$,

$$\begin{aligned} \sum_{i=1}^{n-1} \overline{C}_{e_i} &= \sum_{t=0}^{T-1} ((n-1) - S(t)) \\ &\geq \sum_{t=0}^{n-1} (n-1) \left(1 - \frac{t}{n}\right) \\ &= \frac{(n-1)(n+1)}{2}. \end{aligned}$$

An optimal symmetric LP solution. For every $a \in \bigcup_v G_v$, set

$$x_{a,t} = \frac{1}{nd} \quad (t = 1, \dots, nd).$$

For every helper u_v and every weighted job e_i , set

$$x_{u_v,t} = x_{e_i,t} = \begin{cases} 1/n, & 1 \leq t \leq n, \\ 0, & n < t \leq nd. \end{cases}$$

The nd unit-time jobs contribute exactly one unit of capacity in every slot. For every group G_v ,

$$\sum_{a \in G_v} F_a(t) \geq \min \left\{ \frac{t}{n}, 1 \right\} = F_{u_v}(t),$$

and

$$F_{e_i}(t) = F_{u_0}(t) = F_{u_i}(t).$$

Thus the solution is feasible. Its objective value is

$$(n-1) \frac{1}{n} \sum_{t=1}^n t = \frac{(n-1)(n+1)}{2},$$

which matches the preceding lower bound. The solution is therefore optimal.

A legal bad tie order. In the α -point algorithm, let $\alpha = \beta/d$. In the symmetric solution,

$$Q_a(\beta/d) = Q_{u_v}(\beta) = Q_{e_i}(\beta) = \lceil n\beta \rceil$$

for every a , u_v , and e_i . Hence all keys tie for every realization of β .

A permitted tie order places all unit-time jobs in the leaf groups first, then all unit-time jobs in the center group, then all helpers, and finally all jobs e_i . Every OR-predecessor precedes its helper, and every helper precedes its AND-successors, so this is a legal tie resolution. Every e_i then completes at time nd , giving

$$\text{ALG} = nd(n-1).$$

Together with (6),

$$\frac{\text{ALG}}{\text{OPT}} = \frac{nd(n-1)}{(n-1)(n+2)/2} = \frac{2dn}{n+2}. \quad \square$$

Remark. Proposition 4 does not say that every optimal LP solution or every legal tie-breaking performs badly; favorable optimal solutions or tie rules may produce much better schedules. The proposition only establishes that the algorithm, with otherwise unspecified LP selection and tie-breaking, admits executions attaining the full asymptotic factor.

7 Scope and conclusion

The exact result proved here is

$$g(d) = 2d$$

for the completion-time-indexed relaxation (1a)–(1f) with $p_j \in \{0, 1\}$.

The proof does not establish an all-unit lower bound. Replacing the $\binom{m}{d}$ zero-time jobs b_S and the job z by unit-time jobs would alter the capacity constraints and invalidate the displayed fractional solution. A valid all-unit extension would require a fully specified chain length and a new fractional schedule with verified slot capacity.

Likewise, no claim is made here about arbitrary processing times. For $p_j > 1$, the capacity constraints in (1c) no longer describe the formulation studied in this note, and the geometric interval relaxation of [2] is a different linear program. Its exact integrality gap remains a separate question.

References

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